

Hidden keys and repeated biometric records

Can multiple protected records reveal identity when keys remain hidden?

Fresh-key learned attacks: uncertainty remains; no equivalence claim.

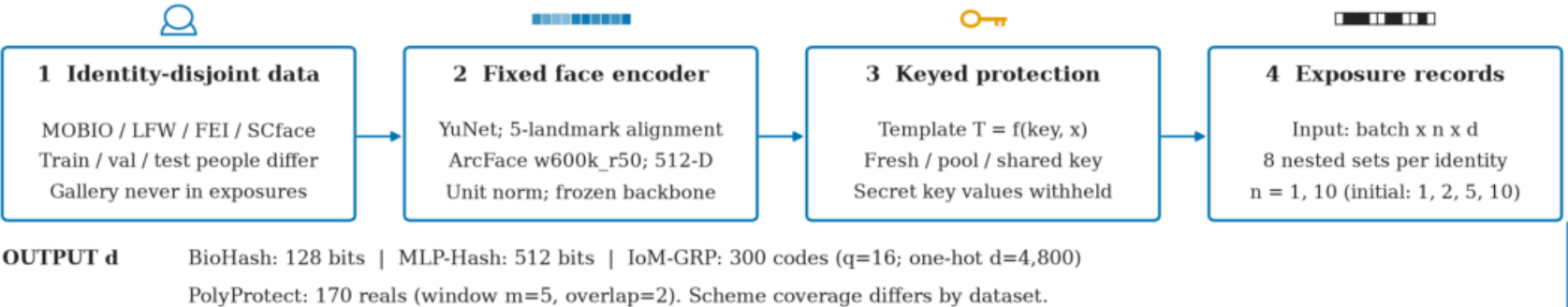
Recurring transforms: large gains from multiple same-identity records.

Pool draws matter; a simple prediction-mean baseline is competitive.

MOBIO/FEI: 216 earlier trained endpoints; 144 new fits plus 72 prediction-mean evaluations.

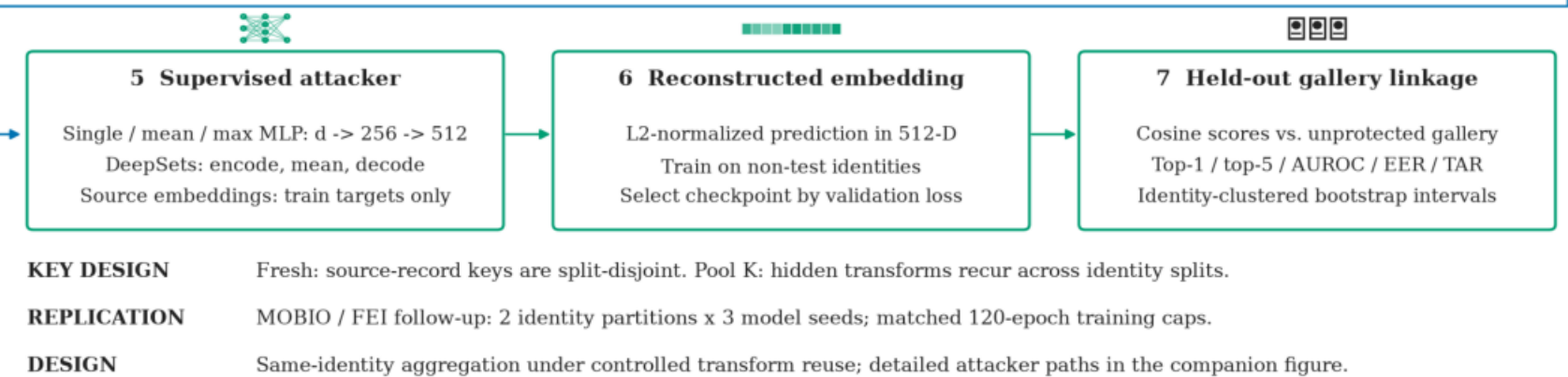
Experimental architecture

A DATA AND PROTECTION



B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



Train on separate identities. Reconstruct an embedding, then link to a held-out gallery.

Dataset and protection coverage

Dataset	Identities	Train/val/test	Embeddings	Variation
MOBIO	150	90/30/30	1,799 / 1,800	Sessions
LFW	125	75/25/25	1,500 / 1,500	In-the-wild
FEI	200	120/40/40	2,378 / 2,400	Pose/expression
SCface	130	78/26/26	2,851 / 2,860	Camera/distance

BioHash: 128 bits; four datasets; includes a Haar-sign control on MOBIO.
MLP-Hash: 512 bits; MOBIO; paper-specified, not source-exact.
IoM-GRP: 300 codes (q=16); PolyProtect: 170 reals (m=5, overlap=2).
Both: MOBIO/FEI three-seed, two-partition follow-up; SCface one-seed pilots.

SCface: mugshot gallery and visible surveillance probes; 9 detection failures, all identities eligible.

Set reconstruction and gallery linkage

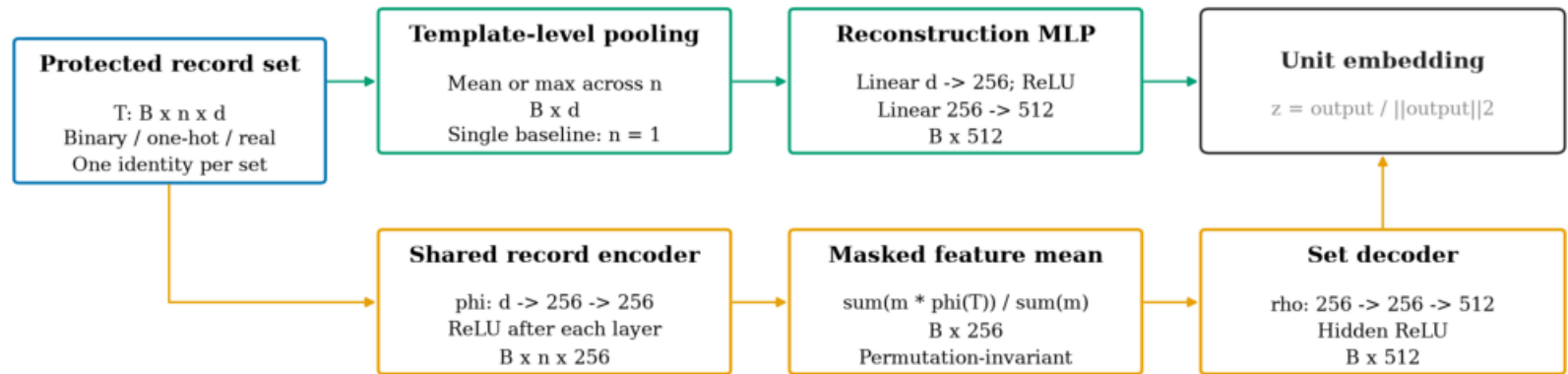
A MATCHED EXPOSURE CONSTRUCTION

One identity, n source records; nested subsets from one fixed permutation per repeat; gallery excluded.

Fresh: independent source-record keys. Pool K: hidden transforms recur across identity splits. No keys or slots in model input.

T: $B \times n \times d$ $d = 128$ BioHash; 512 MLP-Hash; 4,800 one-hot IoM-GRP; 170 PolyProtect

B TWO AGGREGATION PATHS



Primary: mean-pool MLP. Secondary: DeepSets. m masks valid records; all n records are valid in these runs.

C SUPERVISION

Target $u = \text{normalize}(\text{mean of exposed source embeddings})$.

Loss = $\text{mean}[1 - \cos(z, u)] + 0.1 \times \text{MSE}(z, u)$.

Adam: lr 0.001; weight decay 0.0001; hidden width 256.

Train identities only; select minimum validation loss; freeze test.

D HELD-OUT IDENTITY LINKAGE

Scores = $z \cdot G^T$; G: $N \times 512$ unprotected unit gallery.

Rank identities; top-1/top-5, AUROC, EER, TAR at FAR.

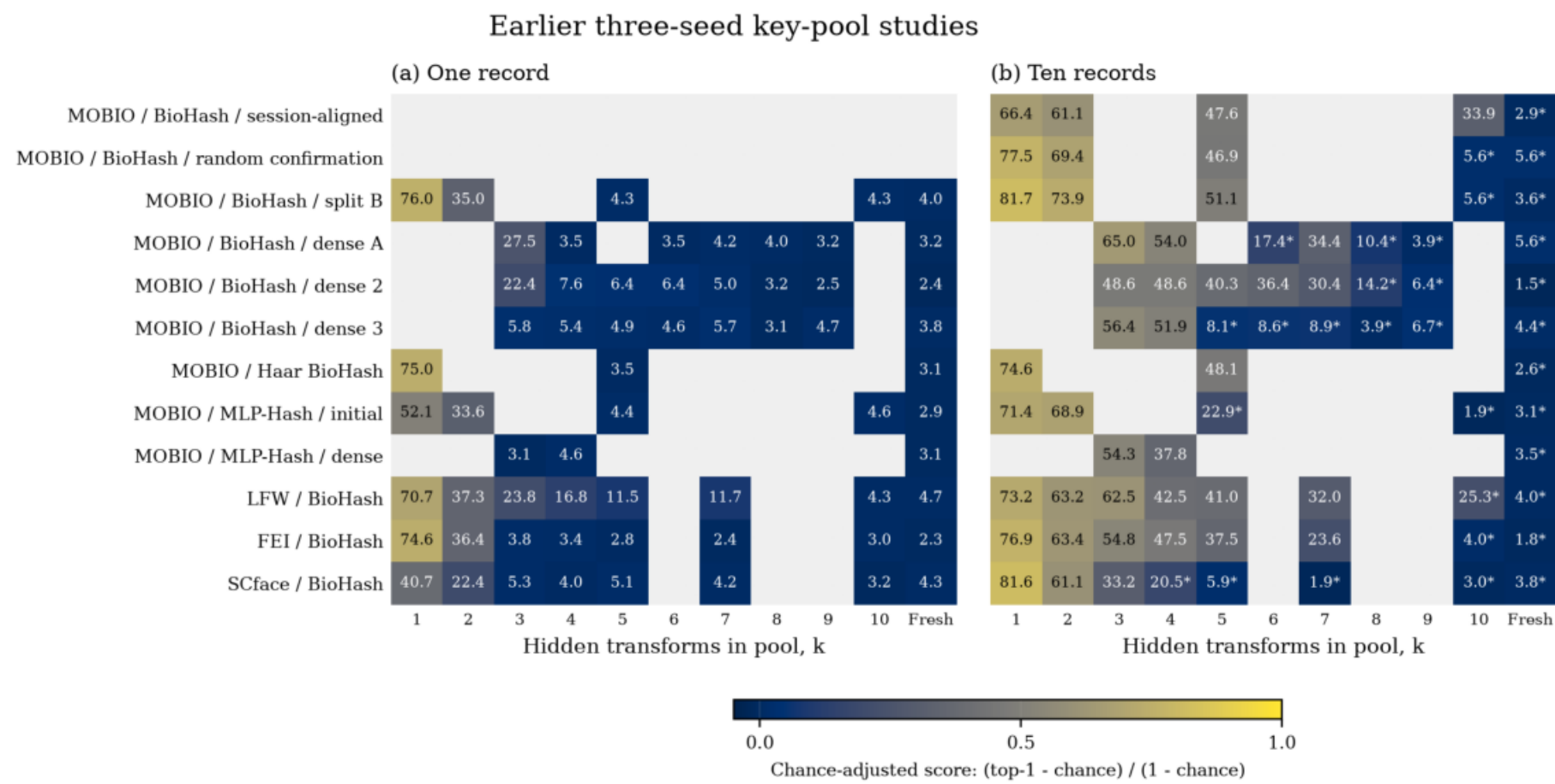
Follow-up: 3 model seeds x 2 identity partitions; 120 epochs.

Crossed seed/identity intervals; paired primary contrasts.

B: batch size; n: exposure count; d: template input width; N: test gallery identities. Source targets are not inference inputs.

Two aggregation paths; shared training objective; held-out identity linkage. Key values remain hidden.

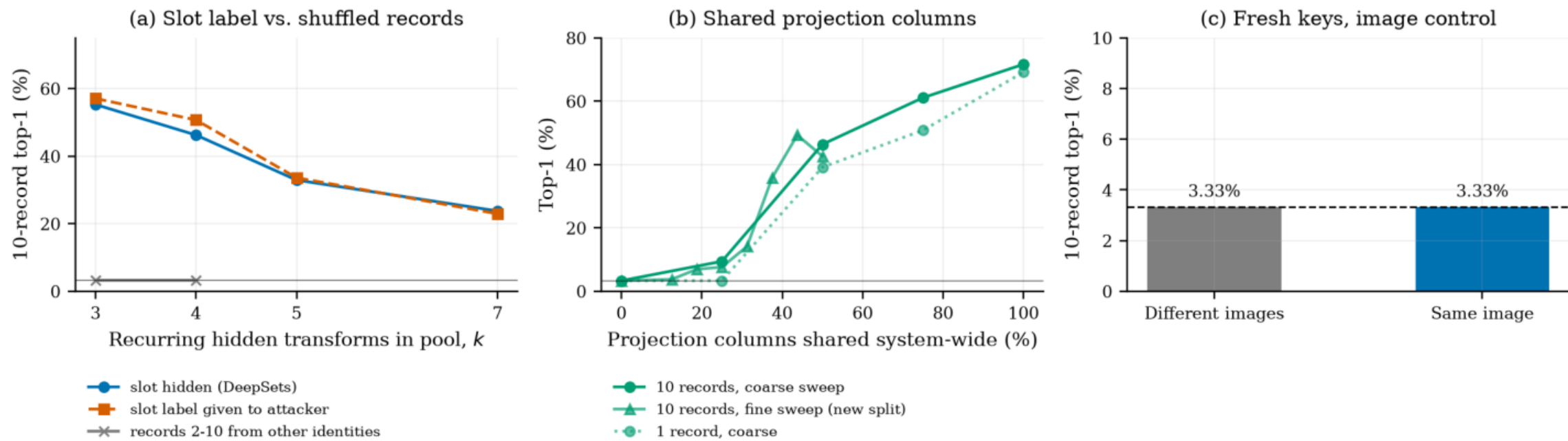
Earlier three-seed key-pool studies



73 conditions from 12 earlier studies. New one-seed pilots are reported separately.

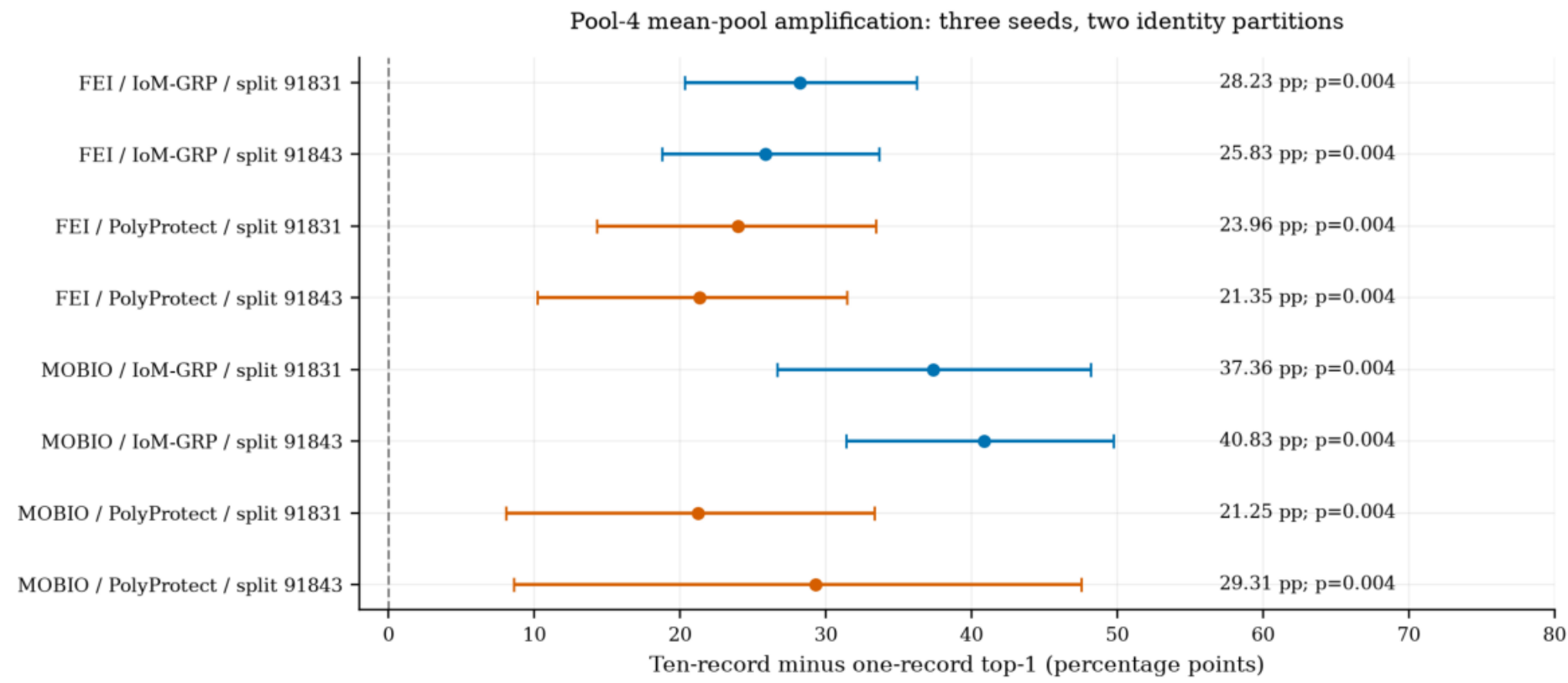
Mechanism and correlation controls

MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Slot identifiers are not key values. Coarse and fine correlation sweeps use separate partitions.

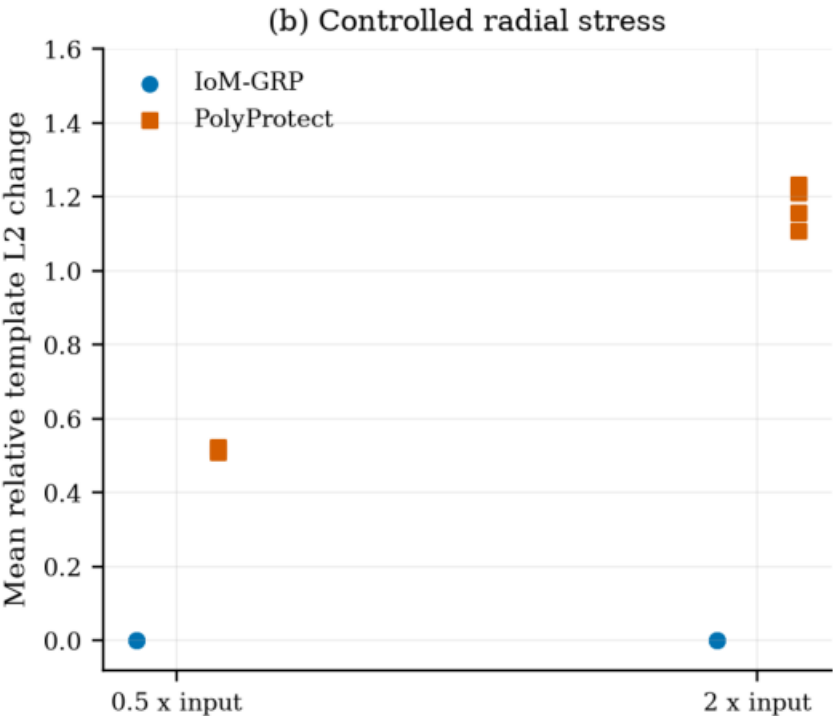
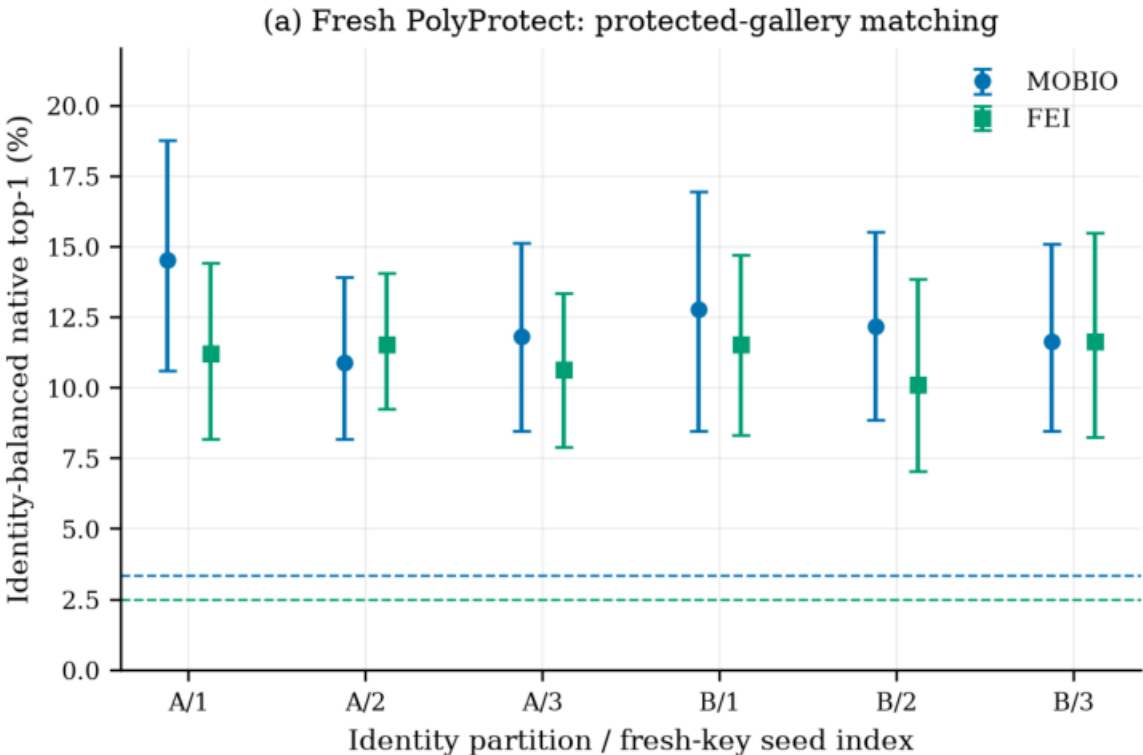
Replicated multi-record amplification



Bars: crossed model-seed / identity bootstrap 95% intervals. Tests: identity sign-flips, Holm family of 8.
Matched 120-epoch caps; key/set seeds fixed. Partitions are separate sensitivity analyses, not independent replications.

MOBIO / FEI; IoM-GRP / PolyProtect; 3 model seeds x 2 identity partitions; matched 120-epoch caps.

Native linkage and radial sensitivity



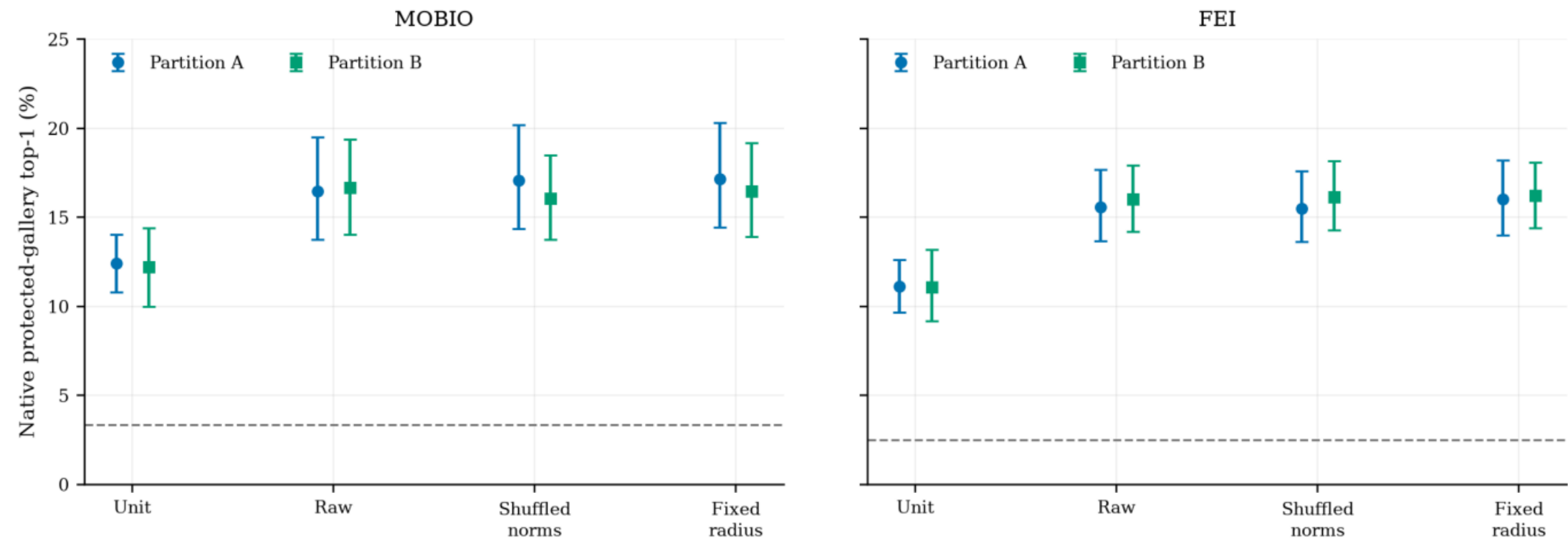
Native tests: 12 / 12 Holm-adjusted $p \leq 0.006$; identity-clustered 95% intervals; dashed lines: chance.

A/B: splits 91831/91843. Key seeds 1/2/3: 91873/91879/91883. Gallery-label permutations preserve probe clusters.

Radial stress: 32 held-out records per dataset/partition, fixed keys; measures scale sensitivity, not natural norm leakage.

Protected-gallery matching is distinct from learned linkage. Radial sensitivity does not establish natural norm leakage.

Genuine raw-input controls



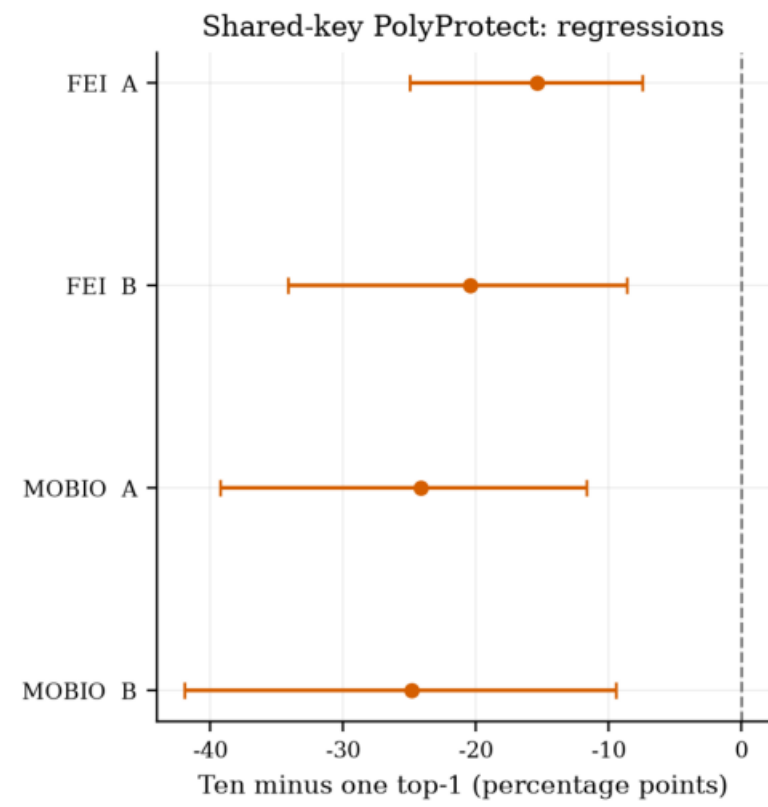
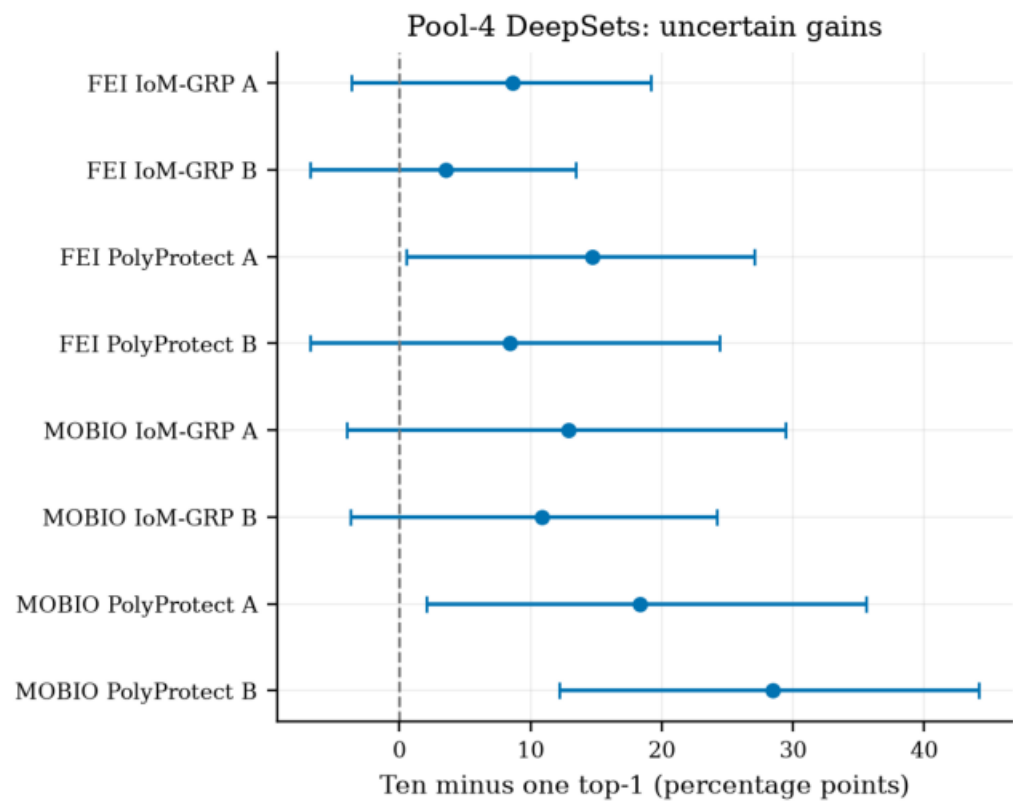
Three fixed key seeds averaged; identity-bootstrap 95% intervals. All 16 null tests: Holm $p = 0.0032$.

Raw minus unit survives paired correction only on FEI. Raw minus shuffled: all adjusted $p \geq 0.7584$.

Natural identity-specific norm leakage is not established. No learned raw-input attack was retrained.

4,177 raw extractions verified. Separate scalar and SciPy checks reproduce native matching; stronger parameter policies remain untested.

Failure analysis: aggregation is not always better



All 48 contrasts reported; post-hoc two-sided Holm family 48. Pointwise crossed seed/identity 95% intervals.

Left: none survives correction (minimum $p = 0.0576$). Right: all four negative contrasts survive ($p = 0.0192$).

All eight pool-4 mean-pool gains remain significant ($p = 0.0192$). More records do not universally improve linkage.

Original primary family unchanged. All 48 contrasts, including significant losses, are reported separately from historical exploration.

Four datasets, distinct evidence layers

Dataset coverage: evidence layers are not interchangeable

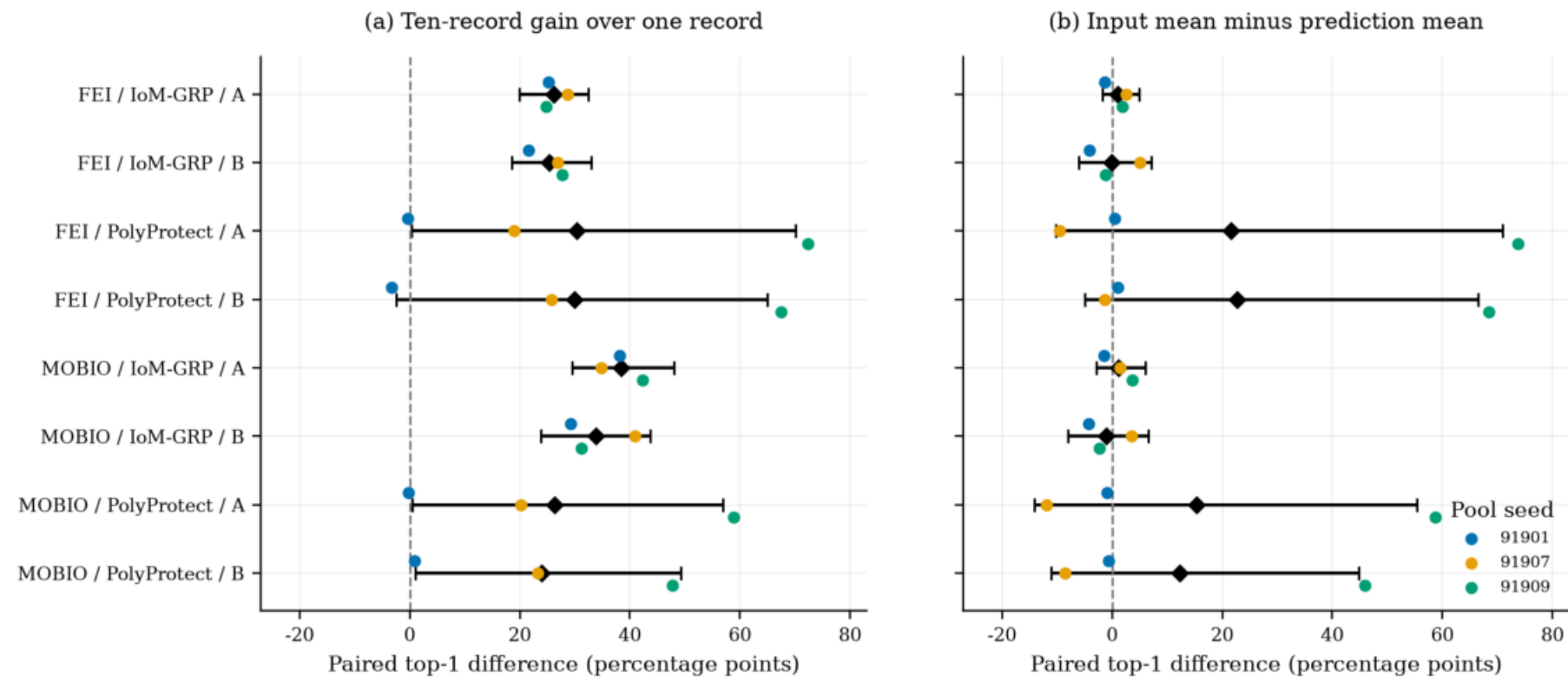
BioHash historical	3 model seeds	3 model seeds	3 model seeds	3 model seeds
MLP-Hash historical	3 model seeds	Not run	Not run	Not run
IoM / PolyProtect pilots	1 model seed	Not run	1 model seed	1 model seed
IoM / PolyProtect follow-up	3 seeds / 2 splits	Not run	3 seeds / 2 splits	Not run
Native / raw-norm audit	3 keys / 2 splits	Not run	3 keys / 2 splits	Not run
Independent-pool + baseline	3 pools / 3 seeds 2 splits	Not run	3 pools / 3 seeds 2 splits	Not run
	MOBIO	LFW	FEI	SCface

SCface retains mugshot-to-surveillance historical and pilot evidence; no new matched confirmation.

Grey = not run in this package, not zero accuracy. Identity partitions overlap; native audits do not train attackers.

SCface remains supporting cross-camera evidence, not matched new-scheme confirmation. Missing cells are not zero accuracy.

Independent pools and a matched baseline



Colored dots: individual pool draws, averaged over three model seeds. Black: mean and crossed pool/seed/identity 95% interval.

IoM gains persist across draws; PolyProtect is pool-sensitive. Every direct baseline-comparison interval includes zero.

Three pools only; pointwise uncertainty, no corrected significance claim. A/B: overlapping partitions 91831/91843.

Three new pool draws: IoM gains persist; PolyProtect is variable. Input-pooling superiority is not established.